

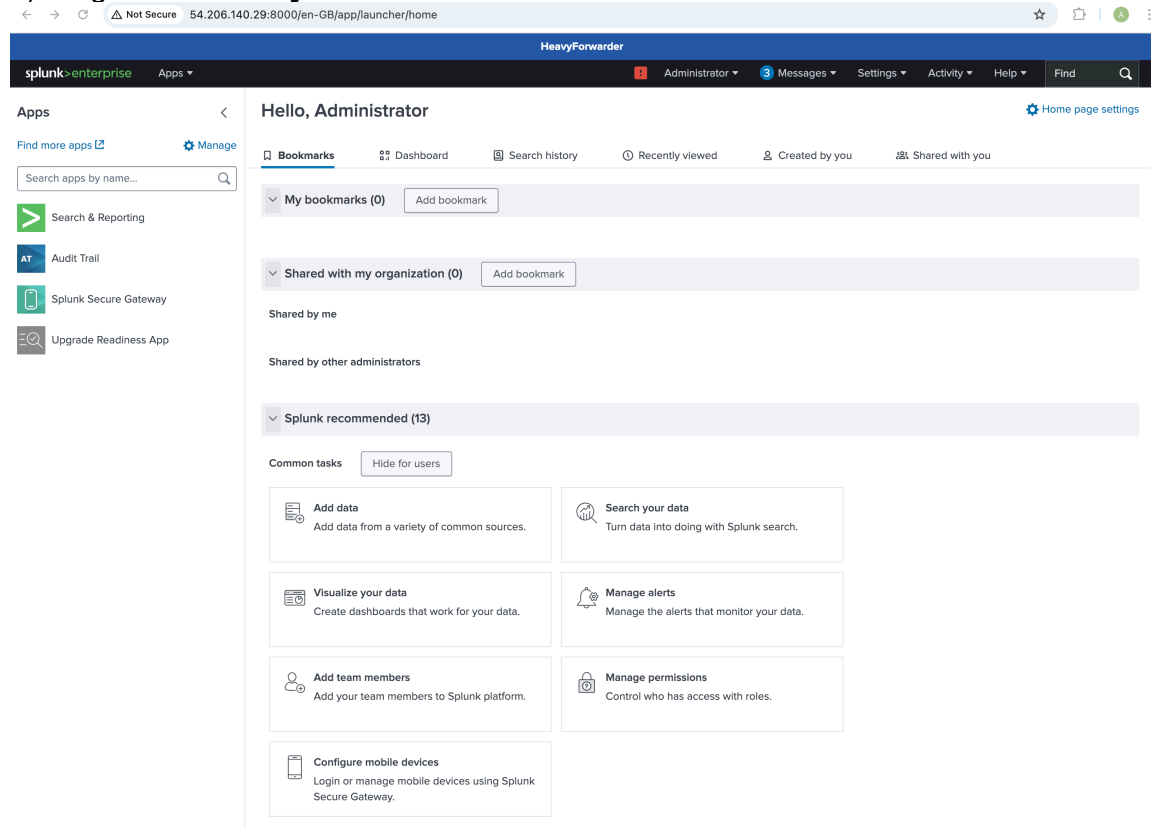
Connect the Universal Forwarder of Linux Server with the Heavy (Intermediate) Forwarder

Summary

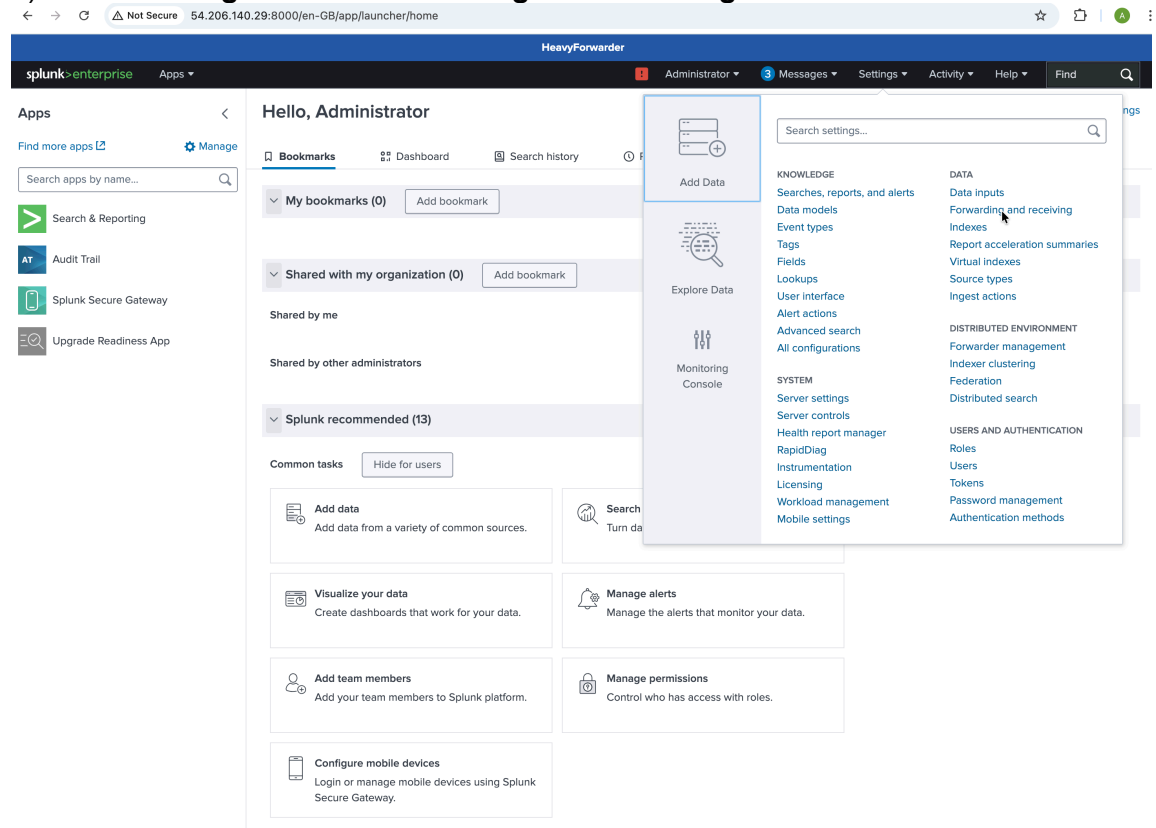
This is a small document describing the steps to connect the Universal Forwarder installed on the Linux Server with the Heavy (Intermediate) Forwarder.

1. Enable Receiving Port in Intermediate Forwarder

1) Login to the HeavyForwarder UI

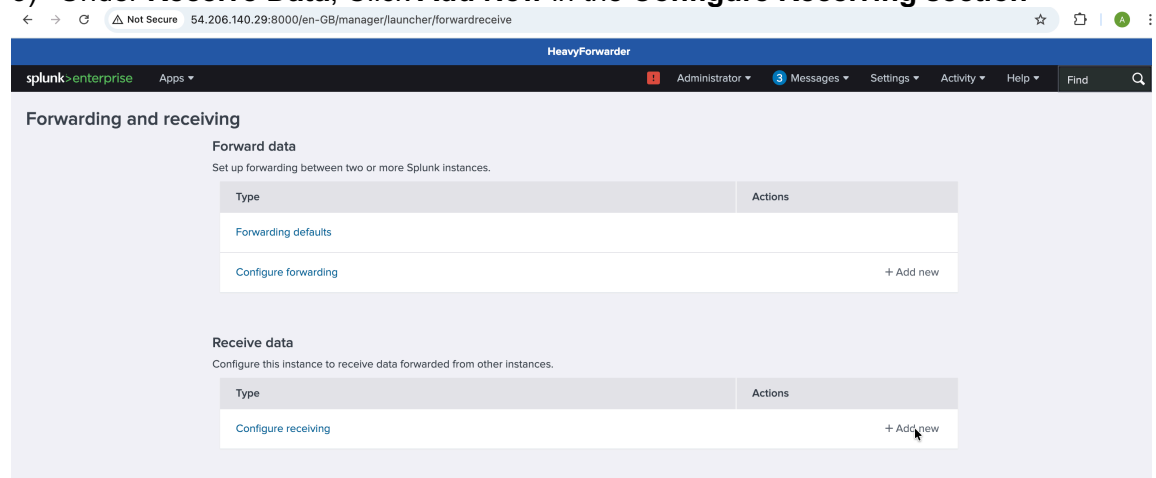


2) Go to **Settings > Data: Forwarding and Receiving**



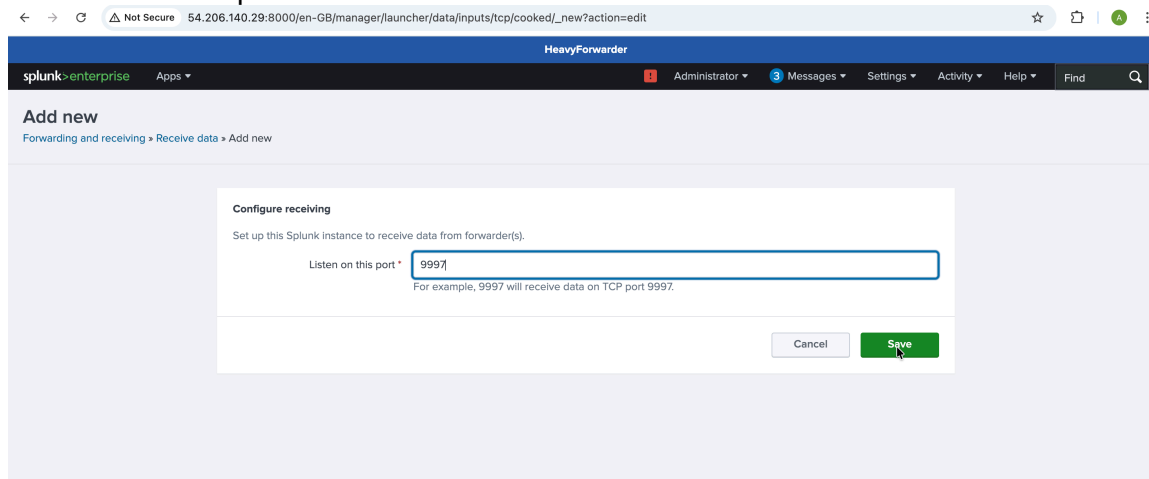
The screenshot shows the Splunk HeavyForwarder interface. On the left, there's a sidebar with 'Apps' and a search bar. The main content area is titled 'Hello, Administrator' and shows a 'Bookmarks' section. A 'Search settings...' dropdown menu is open, displaying a list of settings categories. The 'DATA' category is expanded, and 'Forwarding and receiving' is selected. Other categories visible include KNOWLEDGE, SYSTEM, and DISTRIBUTED ENVIRONMENT.

3) Under **Receive Data**, Click **Add New** in the **Configure Receiving** section



The screenshot shows the 'Forwarding and receiving' settings page. It has two main sections: 'Forward data' and 'Receive data'. The 'Forward data' section has a table with 'Forwarding defaults' and 'Configure forwarding'. The 'Receive data' section has a table with 'Configure receiving'. In the 'Receive data' table, the 'Add new' button is highlighted.

4) In the **New Receiving Port Page**,
Listen on this port: **9997** > **Save**



2. Configure outputs.conf in Universal Forwarder

1) Switch to the splunk user

```
sudo su - splunk
```

2) Navigate to the local configuration directory

```
cd /opt/splunkforwarder/etc/system/local
ls
```

```
[ec2-user@LinuxServer ~]$ sudo su - splunk
Last login: Tue Apr 7 01:23:28 UTC 2026 on pts/0
[splunk@LinuxServer ~]$ cd /opt/splunkforwarder/etc/system/local
[splunk@LinuxServer local]$ ls
README  server.conf
[splunk@LinuxServer local]$ ll
total 8
-r--r--r--. 1 splunk splunk 265 Feb 20 22:23 README
-rw-----. 1 splunk splunk 436 Apr 7 01:08 server.conf
[splunk@LinuxServer local]$
```

3) Create a new file named outputs.conf

```
vi outputs.conf
```

4) Enter the below lines:

```
[tcpout]
defaultGroup=my_Intermediate_Forwarders

[tcpout:my_Intermediate_Forwarders]
server=<HF_PrivateIP>:9997

[tcpout-server://<HF_Private_IP>:9997]
~
~
```

Esc + :wq!

Save and exit the file

```
[tcpout]
defaultGroup=my_Intermediate_Forwarders

[tcpout:my_Intermediate_Forwarders]
server=172.31.0.83:9997

[tcpout-server://172.31.0.83:9997]

~
~
~
~
:wq!
```

5) Exit splunk user and Login as root user to restart Splunk

```
exit
sudo su
cd /opt/splunkforwarder/bin
./splunk restart
```

OR

```
sudo su
/opt/splunkforwarder/bin/splunk restart
```

```
[ec2-user@LinuxServer ~]$ sudo su
[root@LinuxServer ec2-user]# cd /opt/splunkforwarder/bin
[root@LinuxServer bin]# ./splunk restart
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.

Stopping splunk helpers...

Done.
splunkd.pid doesn't exist...

Splunk> The IT Search Engine.

Checking prerequisites...
  Checking mgmt port [8089]: open
  Checking conf files for problems...
  Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4.9-03b
b451d4e07-linux-amd64-manifest'
  All installed files intact.
  Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

[root@LinuxServer bin]#
```

The Universal Forwarder is now configured to forward data to the Intermediate Forwarder.

3. Verify the Connection

On the **SearchHead UI**, open the **Search and Reporting app**
Run the following search:

```
index="_internal"
```

Note: If the search results include logs from the **Universal Forwarder's IP address**, it confirms that the Universal Forwarder is successfully connected to the Intermediate Forwarder.

The screenshot shows the Splunk SearchHead UI with a search for `index="_internal"` returning 16,720 events. A modal window titled "host" is open, showing a table of values for the field "host".

Values	Count	%
Indexer	5,999	35.879%
SearchHead	5,538	33.122%
HeavyForwarder	4,047	24.204%
LinuxServer	1,136	6.794%

The modal also shows a "Reports" section with "Top values" and "Events with this field". The "Values" section is currently selected.

4. Next Steps

Onboard data from the Linux EC2 Instance – Refer [07_Onboarding_data_from_Linux_Server_V1.pdf](#).